PAVAN RAVITEJA UPADHYAYULA

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Professional Summary

Master's student in Cybersecurity at Universität des Saarlandes, researching user privacy perceptions and their mental models as part of my master's thesis. Background in empirical research methods and software security. Currently seeking PhD opportunities in the domain of Usable Security & Privacy, with a primary focus on socio-technical issues in the Global South.

Education

MSc Cybersecurity 04/2023 – Present
Universität des Saarlandes, Germany

- Thesis submission: Nov 2026
- Current grade: 2,3
- Relevant coursework: Human-Computer Interaction, Research methods in HCI, Usable security
- Santander-Stipendium | Stipendienprogramm UdS International - Sommersemester 2026

BTech Computer Science 06/2018 – 04/2022
Koneru Lakshmaiah Educational Foundation (KL University), India

- Final grade: 8.53/10 (German grade: 1,7)

Skills

- **Empirical Methods:** Survey design, mixed-methods analysis, non-parametric testing, thematic analysis
- **Tools:** R, Python, ATLAS.ti, LaTeX, Git, Go
- **Focus Areas:** Usable privacy, Human-Computer Interaction, privacy by design, user mental models, privacy regulation

Research Experience

Master's Thesis Researcher 11/2025 – Present
Internet Architecture (INET) group, MPI-INF, Saarbrücken

- Conducting research on user privacy perceptions of their personal data and its processing in India, in light of its Digital Personal Data Protection Act (DPDPA)
- Designed an exploratory survey (n=239) to understand the perspectives of the educated, English-speaking subset of the Indian population, considered the early adopters of such laws and their provisions
- Analyzed Likert and open-text responses in a mixed-methods design: Friedman & Mann-Whitney U tests, Spearman correlations, and ordinal/linear regression with Holm correction
- Analyzed open-text responses analyzed via reflexive thematic analysis in ATLAS.ti
- Identified the gap between users' privacy concerns, their varied personal data sensitivities, and the provisions of the law expecting uniformity without transparency
- Found near-null demographic effects on users' personal data sensitivity and perceived control over its processing, indicating an underlying distrust towards current regulatory practices

Research Assistant 05/2025 – Present
Fraunhofer IEM, Paderborn (Remote)

- Identified a significant research gap in security metrics and automated risk assessment for software products through a comprehensive literature review
- Evaluated the git repository of an IoT team at DB Systel to help boost their security posture
- Extended the OSSF Scorecard tool (Go) with a new probe measuring pipeline reliability of the main branch
- Applied the extended tool across 25+ repositories; reported concrete security-posture improvements

Publication

P. R. Upadhyayula, K. Amarendra and S. B. Kudupudi, "Intrusion Detection of Imbalanced Network Traffic," *2022 7th International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India, 2022, pp. 840–844, doi: 10.1109/ICCES54183.2022.9835996.

Teaching & Outreach

Student Peer Mentor 04/2019 – 05/2020
KL University, India

- Mentored 300+ peers in object-oriented programming, network security, and software engineering across 10+ interactive, hands-on learning sessions

Public Relations Chair 03/2019 – 11/2021
White Hat Hackers Club, KL University

- Led cybersecurity awareness initiatives, organizing network security workshops for 100+ students and running 4 CTF competitions

Projects

Intrusion Detection of Imbalanced Network Traffic (Bachelor's Thesis)

- Designed and implemented an intrusion detection system leveraging the Modern Random Forest algorithm
- Conducted metadata analysis across multiple KDD datasets, achieving a false alarm rate of ∼5%

Languages

- English: Fluent (C1) | German: Basic (A2)